INTEROPERABILITY PROFILE OF IEC 60870-5-101 PROTOCOL FOR SR

This companion standard presents sets of parameters and alternatives from which subsets have to be selected to implement particular telecontrol systems. Certain parameter values, such as the number of octets in the COMMON ADDRESS of ASDUs represent mutually exclusive alternatives. This means that only one value of the defined parameters is admitted per system. Other parameters, such as the listed set of different process information in command and in monitor direction allow the specification of the complete set or subsets, as appropriate for given applications. This clause summarizes the parameters of the previous clauses to facilitate a suitable selection for a specific application. If a system is composed of equipment stemming from different manufacturers it is necessary that all partners agree on the selected parameters.

The selected parameters should be marked in the white boxes as follows:

☐	Function of ASDU is not used
■	Function or ASDU is used as standardized (default)

Note : In addition, the full specification of a system may require individual selection of certain parameters for certain parts of the system, such as the individual selection of scaling factors for individually addressable measured values.

1.1 SYSTEM or DEVICE

(System-specific parameter, indicate the definition of a system or a device by marking one of the following with an 'X')

- ☐ System definition
- ☐ Controlling station definition (master)
- ■ Controlled station definition (Slave)

1.2 NETWORK CONFIGURATION

(Network-specific parameter, all configurations that are used are to be marked with ■)

- ■ Point-to-point
- ■ Multipoint-party line
- ■ Multiple point-to-point
- ☐ Multipoint-star

1.3 PHYSICAL LAYER (Network-specific parameter)

Transmission speed (control direction) :

Unbalanced interchange circuit V.24/V.28 Standard	*Unbalanced interchange circuit V.24/V.28 Recommended if >1 200 bit/s*	*Balanced interchange circuit X.24/X.27*
☐ 100 bit/s	☐ 2 400 bit/s	☐ 2 400 bit/s
■ 200 bit/s	☐ 4 800 bit/s	☐ 4 800 bit/s
■ 300 bit/s	☐ 9 600 bit/s	☐ 9 600 bit/s
■ 600 bit/s		☐ 19 200 bit/s
■ 1 200 bit/s		☐ 38 400 bit/s
(for unbalanced transmission only)		☐ 56 000 bit/s
		☐ 64 000 bit/s

Transmission speed (monitor direction) :

Unbalanced interchange circuit V.24/V.28 Standard	*Unbalanced interchange circuit V.24/V.28 Recommended if >1 200 bit/s*	*Balanced interchange circuit X.24/X.27*

☐	100 bit/s	☐	2 400 bit/s	☐	2 400 bit/s
■	200 bit/s	☐	4 800 bit/s	☐	4 800 bit/s
■	300 bit/s	☐	9 600 bit/s	☐	9 600 bit/s
■	600 bit/s			☐	19 200 bit/s
■	1 200 bit/s			☐	38 400 bit/s
	(for unbalanced transmission only)			☐	56 000 bit/s
				☐	64 000 bit/s

1.4 LINK LAYER *(Network-specific parameter)*

Frame format FT 1.2, single character 1 and the fixed time out interval are used exclusively in this companion standard.

Link transmission procedure

☐ Balanced transmission

■ Unbalanced transmission

Address field of the link

☐ Not present (balanced transmission only)

■ One octet

☐ Two octets

☐ Structured

■ Unstructured

Frame length

255 Maximum length L (number of octets)

1.5 APPLICATION LAYER

Transmission mode for application data

Mode 1 (Least significant octet first), as defined in clause 4.10 of IEC 870-5-4, is used exclusively in this companion standard.

Common address of ASDU
(system-specific parameter)

■ One octet ☐ Two octets

Information object address
(system-specific parameter)

☐	One octet	☐	Structured
■	Two octets	■	Unstructured
☐	Three octets		

Cause of transmission
(system-specific parameter)

■	One octet	☐	Two octets (with originator address)

Selection of standard ASDUs
(station-specific parameter)

■	<1> :=	Single-point information	M_SP_NA_1
■	<2> :=	Single-point information with time tag	M_SP_TA_1
☐	<3> :=	Double-point information	M_DP_NA_1
☐	<4> :=	Double-point information with time tag	M_DP_TA_1
☐	<5> :=	Step position information	M_ST_NA_1
☐	<6> :=	Step position information with time tag	M_ST_TA_1
■	<7> :=	Bitstring of 32 bit	M_BO_NA_1
☐	<8> :=	Bitstring of 32 bit with time tag	M_BO_TA_1
■	<9> :=	Measured value, normalized value	M_ME_NA_1
☐	<10> :=	Measured value, normalized value with time tag	M_ME_TA_1
☐	<11> :=	Measured value, scaled value	M_ME_NB_1
☐	<12> :=	Measured value, scaled value with time tag	M_ME_TB_1
☐	<13> :=	Measured value, short floating point value	M_ME_NC_1
☐	<14> :=	Measured value, short floating point value with time tag	M_ME_TC_1
■	<15> :=	Integrated totals	M_IT_NA_1
☐	<16> :=	Integrated totals with time tag	M_IT_TA_1
☐	<17> :=	Event of protection equipment with time tag	M_EP_TA_1
☐	<18> :=	Packed start events of protection equipment with time tag	M_EP_TB_1
☐	<19> :=	Packed output circuit information of protection equipment with time tag	M_EP_TC_1

☐	<20> :=	Packed single-point information with status change detection	M_PS_NA_1
☐	<21> :=	Measured value, normalized value without quality descriptor	M_ME_ND_1

Process information in control direction
(station-specific parameter)

■	<45> :=	Single command	C_SC_NA_1
☐	<46> :=	Double command	C_DC_NA_1
☐	<47> :=	Regulating step command	C_RC_NA_1
■	<48> :=	Set point command, normalized value (required only for analog output command)*	C_SE_NA_1
☐	<49> :=	Set point command, scaled value	C_SE_NB_1
☐	<50> :=	Set point command, short floating point value	C_SE_NC_1
☐	<51> :=	Bitstring of 32 bit	C_BO_NA_1

System information in monitor direction
(station-specific parameter)

■	<70> :=	End of initialization	M_EI_NA_1

System information in control direction
(station-specific parameter)

■	<100> :=	Interrogation command	C_IC_NA_1
■	<101> :=	Counter interrogation command	C_CI_NA_1
☐	<102> :=	Read command	C_RD_NA_1
■	<103> :=	Clock synchronization command (optional, if GPS is used for time synch. of the RTU)*	C_CS_NA_1
☐	<104> :=	Test command	C_TS_NA_1
☐	<105> :=	Reset process command	C_RP_NA_1
■	<106> :=	Delay acquisition command (optional, if GPS is used for time synch. of the RTU)*	C_CD_NA_1

Parameter in control direction
(station-specific parameter)

☐	<110> :=	Parameter of measured value, normalized value	P_ME_NA_1
☐	<111> :=	Parameter of measured value, scaled value	P_ME_NB_1
☐	<112> :=	Parameter of measured value, short floating point value	P_ME_NC_1
☐	<113> :=	Parameter activation	P_AC_NA_1

File transfer *(for downloading of database from RLDC, may not be required)* *
(station-specific parameter)

■	<120> :=	File ready	F_FR_NA_1
■	<121> :=	Section ready	F_SR_NA_1
■	<122> :=	Call directory, select file, call file, call section	F_SC_NA_1
■	<123> :=	Last section, last segment	F_LS_NA_1
■	<124> :=	Ack file, ack section	F_AF_NA_1
■	<125> :=	Segment	F_SG_NA_1
☐	<126> :=	Directory	F_DR_TA_1

1.6 BASIC APPLICATION FUNCTIONS

Station initialization
(station-specific parameter)

■ Remote initialization

General interrogation
(system or station-specific parameter)

■ Global

■ Group 1	■ Group 7	■ Group 13
■ Group 2	■ Group 8	■ Group 14
■ Group 3	■ Group 9	■ Group 15
■ Group 4	■ Group 10	■ Group 16
■ Group 5	■ Group 11	
■ Group 6	■ Group 12	Addresses per group have to be defined

Clock synchronization

(station-specific parameter)

■ Clock synchronization (optional, if GPS is used for time synch. of the RTU)*

Command transmission (Required only when control command is envisaged)*
(object-specific parameter)

■ Direct command transmission
■ Direct set point command transmission
■ Select and execute command
■ Select and execute set point
☐ C_SE ACTTERM used

■ No additional definition
☐ Short pulse duration (duration determined by a system parameter in the outstation)
☐ Long pulse duration (duration determined by a system parameter in the outstation)
☐ Persistent output

Transmission of integrated totals
(station or object-specific parameter)

■ Counter request
■ Counter freeze without reset
■ Counter freeze with reset
☐ Counter reset
☐ General request counter
☐ Request counter group 1
☐ Request counter group 2
☐ Request counter group 3
☐ Request counter group 4

Addresses per group have to be defined

Parameter loading
(object-specific parameter)

☐ Threshold value
☐ Smoothing factor
☐ Low limit for transmission of measured value
☐ High limit for transmission of measured value

Parameter activation
(object-specific parameter)

☐ Act/deact of persistent cyclic or periodic transmission of the addressed object

File transfer
(station-specific parameter)

☐ File transfer in monitor direction

■ File transfer in control direction ***(For downloading of database from RLDC, May not be required)****

2. ADDITIONAL INFORMATION ON IEC 60870-5-101 FOR SRLDC

A. Telemetred Data and ASDU mapping

The following table explains the type of the telemetred data and corresponding ASDUs used to transmit this data as per IEC 60870-5-101 protocol

Type of power system Data	Data Unit type as per IEC	Description as per IEC	Data polling method	Interrogation group	Transmitted after Class-X request	Info Obj. Address range
Analog inputs (P,Q, V, f)	ASDU-9	Measured value normalized value	By periodic Group scan	Group-2	Class 2	2001-3000
Digital inputs –Single status (Circuit breakers, Isolators, Protection signals)	ASDU-1	Single point information	By exception (spontaneous) and on periodic Group scan	Group-1	Class 1 after exception, Class 2 after Group 1 scan	1-1000
	ASDU- 2 (for SOE)	Single point information with time tag	By exception (spontaneous)		Class 1 after exception	1001-2000
Pulse accumulators	ASDU-15	Integrated totals	By periodic counter interrogation	Group-1 (counter interrogation)	Class 2	4001-5000
Analog Outputs (Setpoint)	ASDU-48	Set point command Normalized value				5001-6000
Digital Control command (CB Trip/Close)	ASDU 45	Single command				3001-4000

B. DATA POLLING METHOD

1. The RTU shall respond to the Master stations request for the at least the following commands as per the protocol:
   - Status of Link
   - Reset of Link
   - Delay acquisition command *
   - Clock synchronization command *
   - General interrogation command

- Interrogation of Scan group 1 command (all status data)
- Interrogation of Scan group 2 command (all analog data)
- Class 1/2 data polling

If supervisory control commands are envisaged, then SBO procedure is to be used.

2. Normal data polling is by Scan groups
3. All digital inputs are assigned to Scan group-1 and all Analog values are assigned to Scan group-2
4. Analog values are acquired periodically by using the Scan group-2 polling. This periodicity is ranging from 10-15 seconds based on the quantity of analogs and the communication channel bandwidth.
5. Digital input state changes are reported spontaneously by RTU as class 1 data and a integrity scan is performed for all the digital inputs using Scan group-1 at every 10 minutes interval.
6. Double bit digital status data are to be sent as two single-point information from the RTU.

* ***These features may not be required***